ASE American Society of Echocardiography
Heart & Circulation Ultrasound Specialists

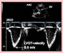
# Echocardiographic Assessment of Valve Stenosis

EUROPEAN ASSOCIATION OF Echocardiography
A Registered Branch of the ESC

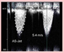
EUROPEAN SOCIETY OF CARDIOLOGY®

## Aortic Stenosis

Left ventricular outflow tract diameter measured within 0.5 to 1.0 cm of the valve orifice.




Left ventricular outflow tract velocity as measured from the apical approach.




Continuous wave Doppler of severe aortic stenosis jet.

### Data recording and measurement for AS quantitation

| Data element | Recording | Measurement |
|---|---|---|
| LVOT diameter | • 2D parasternal long axis view.<br>• Zoom mode.<br>• Adjust gain to optimize the blood tissue interface. | • Inner edge to inner edge.<br>• Mid-systole.<br>• Parallel and adjacent to the aortic valve or at the site of velocity measurement (see text).<br>• Diameter is used to calculate a circular CSA. |
| LVOT velocity | • Pulsed wave Doppler.<br>• Apical long axis or 5-chamber view.<br>• Sample volume positioned just on LV side of valve and moved carefully into the LVOT if required to obtain laminar flow curve.<br>• Velocity baseline and scale adjusted to maximize size of velocity curve.<br>• Time axis (sweep speed) 100 mm/s.<br>• Low wall filter setting.<br>• Smooth velocity curve with a well defined peak and a narrow velocity range at peak velocity. | • Maximum velocity from peak of dense velocity curve.<br>• VTI traced from modal velocity. |
| AS jet velocity | • CW Doppler (dedicated transducer).<br>• Multiple acoustic windows (e.g. apical, suprasternal, right parasternal, etc).<br>• Decrease gains, increase wall filter, adjust baseline and scale to optimize signal.<br>• Gray scale spectral display with expanded time scale.<br>• Velocity range and baseline adjusted so velocity signal fits but fills the vertical scale. | • Maximum velocity at peak of dense velocity curve.<br>• Avoid noise and fine linear signals.<br>• VTI traced from outer edge of dense signal curve.<br>• Mean gradient calculated from traced velocity curve.<br>• Report window where maximum velocity obtained. |
| Valve anatomy | • Parasternal long and short axis views.<br>• Zoom mode. | • Identify number of leaflets in systole, raphe if present.<br>• Assess leaflet mobility and commissural fusion.<br>• Assess valve calcification. |

CSA: cross sectional area, VTI: velocity-time integral.

### Measures of AS severity obtained by Doppler-echocardiography

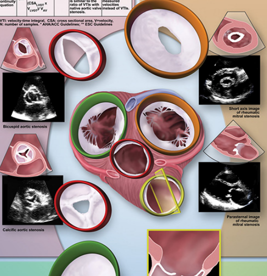
| | Units | Formula / Method | Cutoff for Severe | Concept | Advantages |
|---|---|---|---|---|---|
| AS jet velocity | m/s | Direct measurement | 4.0 | Velocity increases as stenosis severity increase. | Direct measurement of velocity. Strongest predictor of clinical outcome. |
| Mean gradient | mm Hg | $\sum 4v^2 / N$ | 40* or 50** | Pressure gradient calculated from velocity using the Bernoulli equation. | Mean gradient is averaged from the velocity curve. |
| Continuity equation valve area | $cm^2$ | $AVA = (CSA_{LVOT} \times VTI_{LVOT}) / VTI_{AV}$ | 1.0 | Volume flow proximal to and in the stenotic orifice is equal. | Measures effective orifice area. Feasible in nearly all patients. Relatively flow independent. |
| Simplified continuity equation | $cm^2$ | $AVA = (CSA_{LVOT} \times V_{LVOT}) / V_{AV}$ | 1.0 | The ratio of LVOT to aortic velocity is similar to the ratio of VTIs with native aortic valve stenosis. | Uses more easily measured velocities instead of VTIs. |

VTI: velocity-time integral, CSA: cross sectional area, V=velocity, N: number of samples. * AHA/ACC Guidelines; ** ESC Guidelines

### Classification of AS severity

| | Aortic sclerosis | Mild | Moderate | Severe |
|---|---|---|---|---|
| Aortic jet velocity (m/s) | ≤2.5 m/s | 2.6-3.0 | 3-4 | >4 |
| Mean gradient (mm Hg) | | < 20(<30**) | 20-40*(30-50**) | >40* (>50**) |
| AVA ($cm^2$) | | >1.5 | 1.0-1.5 | < 1.0 |
| Indexed AVA ($cm^2/m^2$) | | >0.85 | 0.60-0.85 | < 0.6 |
| Velocity ratio | | >0.50 | 0.25-0.50 | <0.25 |

* AHA/ACC Guidelines; ** ESC Guidelines



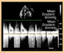
Bicuspid aortic stenosis

Calcific aortic stenosis

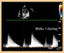
Short axis image of rheumatic mitral stenosis

Parasternal image of rheumatic mitral stenosis

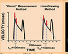
## Mitral Stenosis




Determination of mean mitral gradient from Doppler diastolic mitral flow in a patient with severe mitral stenosis in atrial fibrillation.




Estimation of mitral valve area using the pressure half-time method.




Determination of T1/2 with a bimodal, non-linear slope of the E-wave. The deceleration slope should not be traced from the early part (left), but using the extrapolation of the linear mid-portion of the mitral velocity profile (right).

Planimetry of the mitral orifice. Transthoracic echocardiography, parasternal short-axis view.

### Approaches to evaluation of mitral stenosis

| Measurement | Units | Formula / Method | Concept | Advantages | Disadvantages |
|---|---|---|---|---|---|
| Valve area | | | | | |
| - planimetry by 2D echo | $cm^2$ | tracing mitral orifice using 2D echo | direct measurement of anatomic MVA | - accuracy<br>- independence from other factors | - experience required<br>- not always feasible (poor acoustic window, severe valve calcification) |
| - pressure half-time | $cm^2$ | $220 / T_{1/2}$ | rate of decrease of transmitral flow is inversely proportional to MVA | easy to obtain | dependence on other factors (AR, LA compliance, LV diastolic function) |
| Mean gradient | mm Hg | $\Delta P_{mitral} = \sum 4v^2_{mitral} / N$ | pressure gradient calculated from velocity using the Bernoulli equation | easy to obtain | dependent on heart rate and flow conditions |
| Systolic pulmonary artery pressure | mm Hg | $sPAP = 4v^2_{Tricuspid} + RA_{Pressure}$ | addition of RA pressure and maximum gradient between RV and RA | obtained in most patients with MS | - arbitrary estimation of RA pressure<br>- no estimation of pulmonary vascular resistance |

AR: aortic regurgitation, LA: left atrium, LV: left ventricle, MVA: mitral valve area, ΔP: gradient, sPAP: systolic pulmonary artery pressure, RA: right atrium, RV: right ventricle, $T_{1/2}$: pressure half-time, V=velocity.

### Data recording and measurement in routine use for mitral stenosis quantitation

| Data element | Recording | Measurement |
|---|---|---|
| Planimetry | - determine the smallest orifice by scanning from apex to base<br>- positioning of measurement plan can be oriented by 3D echo<br>- lowest gain setting to visualize the whole mitral orifice | - contour of the inner mitral orifice<br>- include commissures when opened<br>- in mid-diastole (use cine-loop)<br>- average measurements if atrial fibrillation |
| Mitral flow | - continuous-wave Doppler<br>- apical windows often suitable (optimize intercept angle)<br>- adjust gain setting to obtain well-defined flow contour | - mean gradient from the traced contour of the diastolic mitral flow<br>- pressure half-time from the descending slope of the E-wave (mid-diastole slope if not linear)<br>- average measurements if atrial fibrillation |
| Systolic pulmonary artery pressure | - continuous-wave Doppler<br>- multiple acoustic windows to optimize intercept angle | - maximum velocity of tricuspid regurgitant flow<br>- estimation of right atrial pressure according to inferior vena cava diameter |
| Valve Anatomy | - parasternal short-axis view<br><br>- parasternal long-axis view<br><br>- apical two-chamber view | - valve thickness (maximum and heterogeneity)<br>- commissural fusion<br>- extension and location of localized bright zones (fibrous nodules or calcification)<br><br>- valve thickness<br>- extension of calcification<br>- valve pliability<br>- subvalvular apparatus (chordal thickening, fusion, or shortening)<br><br>- subvalvular apparatus (chordal thickening, fusion, or shortening)<br><br>Detail each component and summarize in a score |

### Classification of mitral stenosis severity

| | Mild | Moderate | Severe |
|---|---|---|---|
| Specific findings | | | |
| Valve area ($cm^2$) | > 1.5 | 1.0 - 1.5 | < 1.0 |
| Supportive findings | | | |
| Mean gradient (mm Hg)* | < 5 | 5-10 | > 10 |
| Pulmonary artery pressure (mm Hg) | < 30 | 30-50 | > 50 |

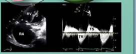
* at heart rates between 60 to 80 beats per minute and in sinus rhythm

## Tricuspid Stenosis

### Findings indicative of hemodynamically significant tricuspid stenosis

| | |
|---|---|
| Specific Findings | |
| Mean pressure gradient | ≥ 5 mm Hg |
| Inflow time velocity integral | > 60 cm |
| $T½$ | ≥ 190 ms |
| Valve area by continuity equation* | ≤ 1 $cm^2$ |
| Supportive Findings | |
| Enlarged right atrium ≥ moderate | |
| Dilated inferior vena cava | |

*Stroke volume derived from left or right ventricular outflow. In the presence of more than mild tricuspid regurgitation, the derived valve area will be underestimated. Nevertheless a value ≤ 1 $cm^2$ implies a significant hemodynamic burden imposed by the combined lesion.



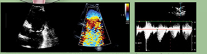

Stenotic tricuspid valve obtained in a modified apical 4-chamber view during diastole.

The peak velocity, diastolic time-velocity integral, mean gradient, and pressure half-time by continuous wave Doppler.

## Pulmonic Stenosis

### Grading of Pulmonary Stenosis

| | Mild | Moderate | Severe |
|---|---|---|---|
| Peak Velocity (m/s) | < 3 | 3-4 | >4 |
| Peak gradient (mm Hg) | < 36 | 36 to 64 | >64 |



Doming of the pulmonary leaflets in systole noted in valvular pulmonary stenosis.

Aliasing of velocities at the level of the pulmonary valve in pulmonary stenosis.

Continuous wave Doppler across the pulmonary valve showing increased velocities.

**Poster ordering information and full text of ASE guideline documents available at: www.asecho.org**

Adapted from: Baumgartner H, Hung J, Bermejo J, Chambers JB, Evangelista A, Griffin BP, Iung B, Otto CM, Pellikka PA, Quiñones M. Echocardiographic Assessment of Valve Stenosis: EAE/ASE Recommendations for Clinical Practice. Eur J Echocardiogr and J Am Soc Echocardiogr 2009.